

TASK # 03 — NETWORK ID & BROADCAST ADDRESSES

Determine the Network ID, Limited Broadcast Address, and Directed Broadcast Address for the following IP addresses.

IP Address	Network ID	Limited Broadcast Address	Directed Broadcast Address
10.0.0.25	10.0.0.0	255.255.255.255	10.255.255.255
192.168.1.100	192.168.1.0	255.255.255.255	192.168.1.255
172.16.5.50	172.16.0.0	255.255.255.255	172.16.255.255
192.168.0.200	192.168.0.0	255.255.255.255	192.168.0.255
150.25.1.10	150.25.0.0	255.255.255.255	150.25.255.255
10.1.2.3	10.0.0.0	255.255.255.255	10.255.255.255
192.168.10.255	192.168.10.0	255.255.255.255	192.168.10.255
172.31.255.255	172.31.0.0	255.255.255.255	172.31.255.255
128.0.0.1	128.0.0.0	255.255.255.255	128.0.255.255
192.0.2.15	192.0.2.0	255.255.255.255	192.0.2.255

TASK # 04 — IP ADDRESS CLASSES

Identify the class (A, B, C, D, or E) of the following IP addresses.

#	IP Address	Class	#	IP Address	Class
1	192.168.1.1	C	11	248.0.0.5	E
2	10.0.0.1	A	12	11.0.0.50	A
3	224.0.0.5	D	13	192.0.2.10	C
4	172.16.5.10	B	14	224.5.5.5	D
5	240.0.0.1	E	15	128.50.10.25	B
6	128.1.0.10	B	16	10.10.10.1	A
7	192.168.0.100	C	17	172.20.1.100	B
8	50.0.0.25	A	18	192.168.5.5	C
9	172.31.255.254	B	19	240.5.5.5	E
10	230.1.1.1	D	20	1.1.1.1	A

TASK # 05 — NETWORK BITS & HOST BITS

Identify the network bits and host bits for the following IP addresses. Highlight the network bits in yellow and the host bits in green.

#	IP Address	Network Bits (yellow) / Host Bits (green)
1	10.0.0.1	
2	192.168.1.1	
3	128.16.5.200	
4	50.25.50.100	
5	172.31.255.255	
6	198.51.100.10	
7	193.0.0.100	
8	100.75.200.25	
9	150.200.100.50	
10	192.168.10.25	

TASK # 09 — VALID HOST RANGE

For the following IP addresses, calculate the second valid host ID and the last valid host ID.

#	IP Address	Second Valid Host ID	Last Valid Host ID
1	10.15.30.45	10.0.0.2	10.255.255.254
2	172.30.100.50	172.30.0.2	172.30.255.254
3	192.168.5.75	192.168.5.2	192.168.5.254
4	8.20.40.60	8.0.0.2	8.255.255.254
5	192.0.2.150	192.0.2.2	192.0.2.254